

SATYA RAJANALA

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PROFESSIONAL SUMMARY

- Master of Science in Computer Science graduate from Kent State University with strong expertise in Python Development, Machine Learning, and Data Analysis.
- Gained hands-on experience through Graduate Assistantship, internships, and research projects, contributing to academic instruction, software development, and applied machine learning.
- Skilled in Python, Flask, Django, NumPy, Pandas, Scikit-learn, SQL, and cloud platforms (AWS, Azure) with proven ability to build and deploy real-world applications.
- Assisted in research across Artificial Intelligence, Blockchain, and Cloud Computing, contributing to code development, data analysis, and technical publications.
- Developed and deployed end-to-end machine learning solutions, including data preprocessing, feature engineering, model evaluation, and Flask-based web applications.
- Recognized for mentoring students, troubleshooting technical environments, and delivering automated solutions to support large-scale academic projects.
- I am passionate about designing scalable applications and eager to contribute to innovative projects in Python development, data science, and cloud computing.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, .NET

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, Flask, Django

Databases: MySQL, MS-SQL Server, Oracle

Web Technologies: HTML, CSS, JavaScript, XML

Cloud Platforms: AWS, Azure

Tools & Others: GitHub, RESTful APIs, Data Analysis & Visualization, Agile Methodologies, CI/CD, Linux/Windows environments

EDUCATION

Kent State University, Kent, OH | Master of Science in Computer Science, Dec 2024

NRI Institute of Technology, India | Bachelor of Science in Computer Science, May 2023

PROFESSIONAL EXPERIENCE

Graduate Student Assistant | Kent State University | Kent, OH | Jan 2024 – Dec 2024

- Guided lab sessions and discussions for courses such as Machine Learning, Algorithms, and Software Engineering.
- Supported students with advanced programming concepts and assisted with troubleshooting software environments and version control.
- Contributed to research in Artificial Intelligence, Blockchain, and Cloud Computing by assisting in code development, data analysis, and paper writing.
- Managed cloud-based services and provided Tier-2 support for technical issues.

Project Intern | NRI Institute of Technology | India | Dec 2022 – May 2023

- Designed and implemented a Network Intrusion Detection System (NIDS) using Python.

- Built supervised ML models (Support Vector Machine, ANN) to classify traffic as malicious or benign.
- Improved model performance with advanced feature selection techniques.
- Documented findings and collaborated with faculty for evaluation and enhancements.

Research Assistant | NRI Institute of Technology | India | Jul 2021 – May 2023

- Developed a Python-based Base Converter application with integrated calculator features.
- Implemented functionality to convert numbers between multiple bases and created an intuitive user interface.
- Applied software development best practices and testing methodologies.

PROJECTS

Air Quality Prediction using Machine Learning | Kent State University | Aug 2024 – Dec 2024

- Implemented a Flask-based machine learning model for Air Quality Index (AQI) prediction using Decision Tree and Random Forest regression.
- Processed pollutant datasets (PM2.5, NOx, CO) to classify air quality categories (Good, Moderate, Poor).
- Evaluated models with RMSE, MAE, and R^2 metrics; Random Forest achieved superior accuracy.
- Built a web-based interface for real-time predictions.

CERTIFICATION

NPTEL Certification in Programming in Java

NPTEL Certification in Cloud Computing